RETAIL SALES PERFORMANCE AND DISCOUNT IMPACT ANALYSIS USING POWER BI

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

In today's data-driven business environment, analyzing sales performance and understanding the impact of discounts on profitability are crucial for effective decision-making. This project, titled **"Retail Sales Performance & Discount Impact Analysis using Power BI"**, focuses on developing an interactive and insightful dashboard to analyze retail sales data and derive meaningful business insights.

The dataset used in this project is sourced from a publicly available and authentic dataset, which includes information such as sales, profit, discount, customer details, product categories, and regional performance. The project involves data preprocessing using Power Query, followed by data modeling and the creation of calculated measures using DAX (Data Analysis Expressions).

The developed dashboard is divided into three main sections: Overview, Discount Analysis, and Customer & Product Insights. The Overview page provides a summary of key performance indicators such as total sales, total profit, and order count. The Discount Analysis page explores the relationship between discount levels and profitability, highlighting how higher discounts can lead to reduced profit margins. The Customer & Product Insights page identifies top-performing customers and products, along with category-wise and segment-wise performance.

The results of this analysis reveal that while discounts can boost sales volume, excessive discounts negatively impact profitability. Additionally, a significant portion of revenue is generated by a small group of customers and products, indicating a concentration of business performance.

This project demonstrates the effectiveness of Power BI as a business intelligence tool for transforming raw data into actionable insights, thereby enabling better strategic decision-making.

GRAPHICAL ABSTRACT

The graphical abstract represents the overall workflow of the project, illustrating the transformation of raw retail data into meaningful business insights. The process begins with the collection of a structured dataset containing detailed information on sales transactions, customers, products, and regions. This data is then prepared and refined through systematic cleaning and transformation techniques to ensure accuracy and consistency. A well-defined data model is subsequently developed to organize relationships and enable efficient analysis. Key performance metrics are computed using advanced expressions to support dynamic evaluation of business performance. The processed data is then visualized through an interactive multi-page dashboard, enabling comprehensive exploration of sales trends, discount effects, customer behavior. The final stage highlights the major analytical outcomes, emphasizing patterns in profitability, purchasing behavior, and category-level performance, thereby supporting informed decision-making development

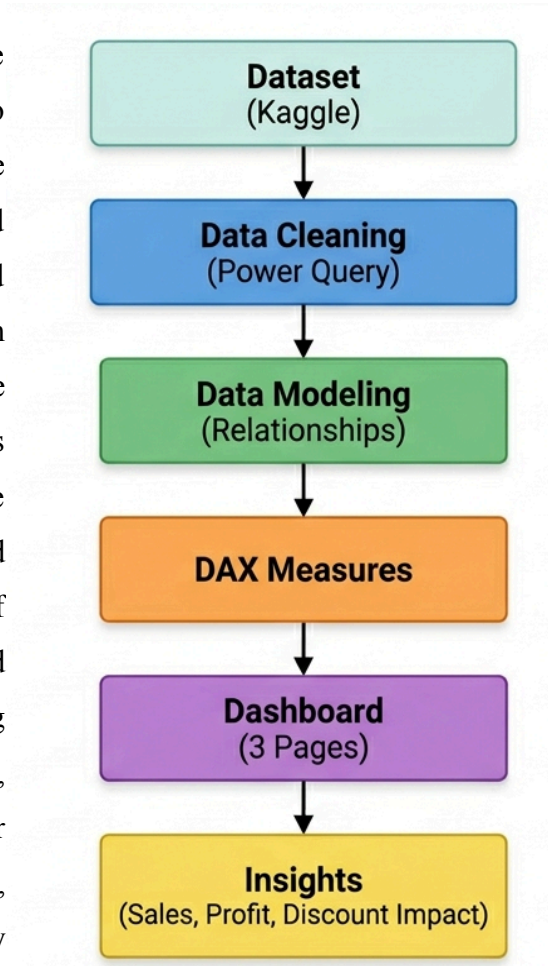

Figure 0.1: Workflow of data analysis & dashboard

1. INTRODUCTION

1.1. Identification of Client /Need / Relevant Contemporary issue

In the modern business environment, organizations generate vast amounts of data through daily operations such as sales transactions, customer interactions, and product management. However, the presence of data alone is not sufficient; it must be analyzed effectively to extract meaningful insights. Retail businesses, in particular, face significant challenges in understanding sales performance, customer behavior, and the impact of pricing strategies such as discounts.

The increasing competition in the retail industry has made it essential for organizations to adopt data-driven decision-making approaches. Without proper analysis, businesses may suffer from reduced profitability, inefficient operations, and poor strategic planning. One of the most critical challenges is understanding how discounts influence both sales and profit. While discounts can attract customers and increase sales volume, they can also lead to a decline in overall profitability if not managed properly.

The need for an interactive and dynamic solution arises from the limitations of traditional reporting systems, which often present static and difficult-to-interpret data. Business managers require tools that can provide real-time insights, visual representation of trends, and the ability to explore data from multiple perspectives.

This project addresses these needs by developing a comprehensive dashboard using Power BI, which enables effective visualization and analysis of retail sales data. The solution helps stakeholders identify key performance indicators, monitor business trends, and make informed decisions. The relevance of this project is aligned with current industry practices where organizations increasingly rely on business intelligence tools to gain a competitive advantage.

1.2. Identification of Problem

Despite the availability of large volumes of retail data, many organizations struggle to convert this data into actionable insights. Traditional methods of data analysis, such as spreadsheets and manual reporting, are time-consuming, error-prone, and lack interactivity.

One of the primary problems in retail analysis is the inability to clearly understand the relationship between discount strategies and business outcomes. Businesses often implement discounts without analyzing their long-term effects on profitability. This can result in situations where increased sales do not necessarily translate into increased profits.

Another significant issue is the lack of a centralized system that integrates various aspects of business data, such as customer details, product performance, regional sales, and financial metrics. Without such integration, it becomes difficult to gain a holistic view of the business.

Additionally, decision-makers often face challenges in identifying top-performing customers and products, which are crucial for revenue generation. The absence of proper visualization tools further complicates the process, as raw data is difficult to interpret without graphical representation.

Therefore, there is a need to design a system that can efficiently process retail data, analyze key factors such as discounts and profitability, and present the results in an intuitive and interactive format.

1.3. Identification of Tasks

To address the identified problems, the project is divided into a series of well-defined tasks that ensure systematic development and implementation of the solution.

The first task involves data collection from a reliable and authentic source. The dataset must contain sufficient information related to sales transactions, customer details, product categories, and discounts.

The second task focuses on data preprocessing, which includes cleaning, transforming, and organizing the dataset. This step is essential to remove inconsistencies, handle missing values, and ensure that the data is suitable for analysis.

The third task involves data modeling, where relationships between different data elements are established. This helps in creating a structured framework that supports efficient querying and analysis.

The fourth task includes the creation of calculated measures using DAX (Data Analysis Expressions). These measures are used to compute key performance indicators such as total sales, total profit, profit margin, and average discount.

The fifth task is the development of an interactive dashboard using Power BI. The dashboard is divided into multiple sections to provide a comprehensive view of the data.

The final task involves analysis and interpretation of results, where insights are derived from the visualizations to support business decision-making.

1.4. Timeline

The project was completed following a structured timeline to ensure systematic progress and timely completion of each phase. The development process was divided into multiple stages, including data collection, preprocessing, modeling, visualization, and analysis.

Initially, time was allocated for understanding the dataset and identifying the key objectives of the project. This was followed by data cleaning and transformation, which required careful handling of inconsistencies and formatting issues.

The next phase involved building the data model and creating DAX measures, which formed the foundation for analysis. Subsequently, the dashboard was developed with multiple pages focusing on different aspects of the business.

Finally, time was dedicated to testing, validation, and refinement of the dashboard to ensure accuracy and usability.

1.5. Organization of the Report

This report is organized into five chapters, each addressing a key aspect of the project.

Chapter 1 introduces the problem and objectives.
Chapter 2 reviews existing studies and solutions.
Chapter 3 explains the design and methodology used.
Chapter 4 presents the results and analysis.
Chapter 5 concludes the report and outlines future enhancements.

2. LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the reported problem

The problem of analyzing retail sales data and understanding customer purchasing behavior has been studied extensively over the past few decades. Initially, retail businesses relied on manual record-keeping and basic statistical methods to track sales performance. These traditional approaches were limited in scope and lacked the ability to process large volumes of data efficiently.

With the advancement of technology in the late 20th century, organizations began adopting computerized systems and databases to store transactional data. This marked the beginning of structured data analysis in retail. However, these systems primarily focused on data storage rather than meaningful analysis.

In the early 2000s, the concept of data warehousing and business intelligence emerged, enabling organizations to analyze historical data and generate reports. Tools such as OLAP (Online Analytical Processing) allowed multidimensional analysis of data, providing deeper insights into sales trends and customer behavior.

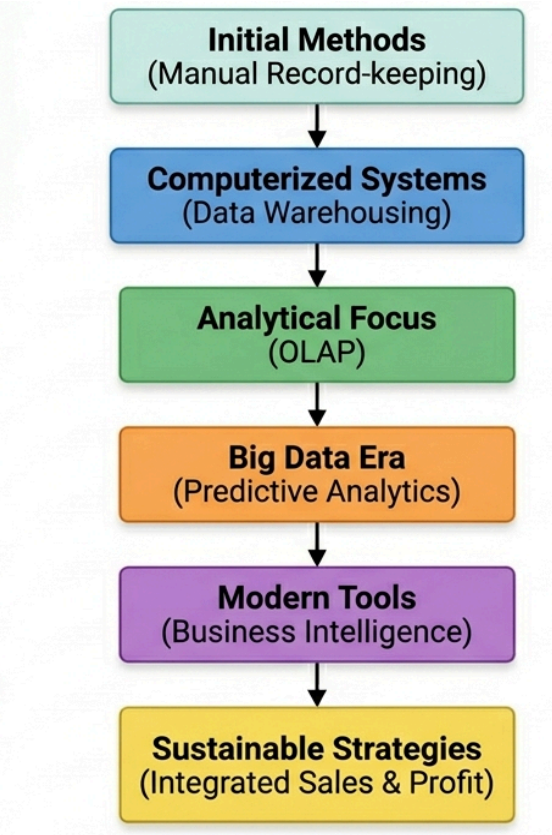

Figure 2.1: Evolution of Retail Data Analysis

In recent years, the rapid growth of big data and analytics has transformed the retail industry. Modern tools such as Power BI, Tableau, and other business intelligence platforms have made it possible to create interactive dashboards and perform real-time analysis. These tools enable businesses to identify patterns, predict trends, and make informed decisions.

The issue of discount impact on profitability has gained significant attention in recent research. Studies indicate that while discounts can attract more customers and increase sales volume, they often lead to a reduction in profit margins if applied excessively. This creates a challenge for businesses to balance sales growth with profitability. Therefore, it is essential to analyze both sales and profit together to ensure sustainable and efficient business performance.

2.2. Existing solutions

Various solutions have been developed over time to address the challenges of retail data analysis. One of the earliest methods involved the use of spreadsheet tools such as Microsoft Excel. Excel provides basic functionalities for data organization, calculations, and visualization through charts and pivot tables. While it is useful for small datasets, it becomes inefficient when dealing with large and complex data.

Another approach involves the use of database management systems (DBMS) such as MySQL and Oracle. These systems allow efficient storage and retrieval of data but require additional tools for visualization and analysis.

Business intelligence tools have emerged as the most effective solution for data analysis in recent years. Tools such as Power BI, Tableau, and QlikView provide advanced capabilities for data visualization, interactive reporting, and real-time analysis. These tools enable users to create dashboards that combine multiple data sources and present insights in an intuitive format.

Among these tools, Power BI is widely used due to its user-friendly interface, integration capabilities, and powerful data processing features. It allows users to perform data transformation, create relationships between datasets, and generate interactive visualizations.

Despite these advancements, existing solutions often face challenges such as data complexity, integration issues, and the need for technical expertise. Therefore, there is a continuous need to improve data analysis techniques and develop solutions that are both efficient and user-friendly.

2.3. Bibliometric analysis

A bibliometric analysis of existing studies reveals that research in retail data analytics has focused on several key areas, including sales forecasting, customer segmentation, and pricing strategies. Many studies emphasize the importance of understanding customer behavior to improve sales performance and enhance customer satisfaction.

Research highlights the use of machine learning to predict sales and analyze customer behavior, but it requires advanced skills and resources.

Another area of focus is the analysis of discount strategies. Studies indicate that excessive discounting can lead to reduced profitability, even if it increases sales volume. Therefore, it is essential to strike a balance between attracting customers and maintaining profit margins.

In terms of visualization, several studies emphasize the importance of dashboards in simplifying complex data. Interactive dashboards enable users to explore data from different perspectives, identify trends, and make informed decisions.

However, many existing studies lack practical implementation or focus only on theoretical aspects. This creates a gap between research and real-world applications, which this project aims to address by developing a practical and interactive solution.

2.4. Review Summary

The literature review highlights the evolution of retail data analysis from basic statistical techniques to advanced business intelligence tools. Modern tools offer significant advantages in terms of efficiency, scalability, and ease of use, enabling organizations to process large volumes of data effectively.

The review also indicates that although multiple solutions are available, there remains a need for integrated systems that combine data processing, analysis, and visualization within a single platform. Power BI emerges as a suitable solution due to its comprehensive features, user-friendly interface, and strong data integration capabilities.

Furthermore, existing studies emphasize the importance of analyzing the relationship between sales, discounts, and profitability. Understanding this relationship is essential for making informed business decisions and ensuring long-term sustainability.

Based on these observations, the project focuses on developing an interactive Power BI dashboard that addresses the limitations of traditional data analysis methods. The proposed solution aims to integrate data processing, visualization, and analysis into a single platform, making it easier to interpret complex data. It provides clear and actionable insights into retail sales performance, helping users make informed and effective business decisions.

2.5. Problem Definition

The primary problem addressed in this project is the lack of an efficient and interactive system for analyzing retail sales data and understanding the impact of discounts on business performance.

The project aims to develop a solution that can process large volumes of data, generate key performance indicators, and provide visual insights into sales trends, customer behavior, and product performance.

The system should be capable of identifying patterns and relationships within the data, such as the effect of discounts on profitability and the contribution of different customer segments to overall sales.

At the same time, the solution should be user-friendly and accessible to non-technical users, enabling them to explore data and make informed decisions without requiring advanced technical knowledge.

The project does not focus on predictive modeling or complex machine learning techniques, but rather on descriptive and analytical reporting using business intelligence tools.

2.6. Goals/Objectives

The main objective of this project is to develop an interactive dashboard that provides insights into retail sales performance. The specific objectives include analyzing sales data, identifying key performance indicators, evaluating the impact of discounts on profitability, recognizing top customers and products, and examining category and segment performance. The project also aims to design a user-friendly Power BI dashboard that supports data-driven decision-making.

3. DESIGN FLOW/PROCESS

3.1. Overview

This chapter describes the overall design and methodology adopted for the development of the Retail Sales Performance & Discount Impact Analysis dashboard using Power BI. The process involves multiple stages including data collection, data preprocessing, data modeling, measure creation, and dashboard development. Each stage is carefully planned and executed to ensure accuracy, efficiency, and meaningful output.

The methodology follows a systematic and structured approach to transform raw data into actionable insights through effective processing and visualization techniques. It emphasizes the importance of data integrity, proper organization, and logical flow of information throughout the system. By following a step-by-step process, the project ensures that each component contributes to the final analytical outcome.

In addition, the methodology focuses on creating a scalable and flexible framework that allows easy modification and future enhancements. The use of Power BI enables seamless integration of data transformation, modeling, and visualization within a single platform. This integrated approach improves efficiency and ensures consistency across all stages of analysis.

Overall, this chapter provides a detailed explanation of how different components work together to convert raw data into an interactive dashboard, enabling users to explore insights, identify trends, and support data-driven decision-making.

3.2. Data Collection

The dataset used in this project is sourced from a publicly available dataset on Kaggle, commonly known as the Global Superstore dataset. This dataset contains detailed information about retail transactions, including sales, profit, discount, customer details, product categories, and geographical information. It provides a comprehensive view of business operations across multiple dimensions.

The dataset is well-structured and consists of multiple attributes that are essential for analyzing business performance. These attributes enable detailed exploration of sales trends, profitability patterns, customer behavior, and regional performance. The availability of diverse fields allows for multi-dimensional analysis, making it suitable for building an interactive and insightful dashboard.

In addition, the dataset spans across different time periods, which makes it useful for performing time-based analysis such as trend identification and seasonal pattern recognition. The inclusion of customer and product-level data further supports detailed segmentation and performance evaluation.

Before analysis, the dataset is imported into Power BI, where it serves as the primary data source for all transformations and visualizations. Its structured format ensures compatibility with data modeling techniques and facilitates efficient data processing.

Table 3.1: Description of Dataset Attributes

Column Name	**Description**
Order Date	Date of purchase
Sales	Total revenue generated
Profit	Not earnings
Discount	Discount applied
Category	Product category
Customer Name	Buyer details
Region	Sales region

The above table provides a summary of the key attributes used in the dataset. These fields form the foundation for analysis and are utilized in creating measures, relationships, and visualizations within the dashboard. Each attribute plays a specific role in understanding different aspects of business performance.

3.3. Data Preprocessing

Data preprocessing is a crucial step that ensures the quality, consistency, and reliability of the dataset before performing analysis. Raw data often contains inconsistencies, missing values, duplicate entries, and formatting issues that can affect the accuracy of results. Therefore, proper preprocessing is essential to prepare the data for effective analysis.

In this project, Power Query is used as the primary tool for data cleaning and transformation. Various preprocessing steps are performed, including the removal of duplicate records to avoid redundancy, handling of missing and null values to maintain data completeness, and correction of data types such as converting columns into appropriate formats like date, numeric, and text.

Additionally, new columns are created to enhance analytical capabilities, such as extracting time-based features (month, year) and categorizing discount values. Column names are also standardized and renamed where necessary to improve clarity and usability within the data model.

These preprocessing steps ensure that the dataset is clean, structured, and ready for further processing. By improving data accuracy and consistency, preprocessing plays a vital role in enabling reliable analysis and meaningful visualization in the dashboard.

3.4. Data Modeling

Data modeling is a critical step in the data analysis process, as it establishes relationships between different tables to enable efficient data organization and retrieval. A well-structured data model ensures that the data is logically arranged, which improves performance and supports accurate analysis.

In this project, a structured data model is created by connecting the main sales table with dimension tables such as customer, product, and date. This approach follows the star schema model, where the central fact table (sales data) is linked to multiple dimension tables. The star schema simplifies data relationships, reduces complexity, and enhances query performance within Power BI.

The relationships between tables are defined using common fields such as customer ID, product ID, and order date. These relationships allow seamless interaction between different datasets and enable the creation of meaningful visualizations. The use of a date table further enhances time-based analysis by supporting features such as monthly and yearly trends.

Proper data modeling also ensures that calculations and measures work accurately across different tables. It improves data consistency and prevents redundancy, making the analysis more reliable and efficient.

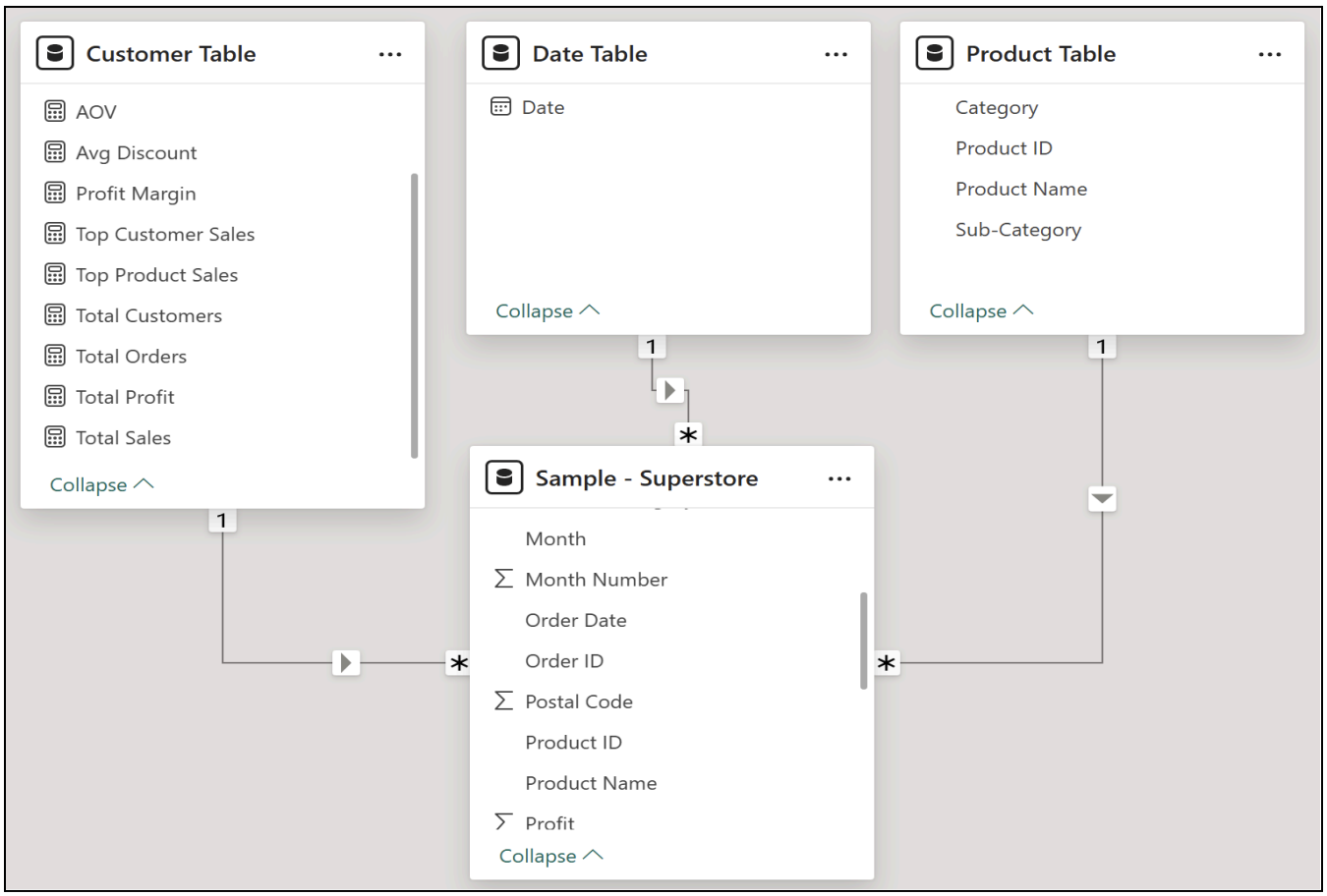

Figure 3.1: Data Model showing relationships between fact and dimension tables

3.5. Design Flow

The overall workflow of the system follows a structured and systematic pipeline, where data flows from raw input to final visualization. This flow ensures that each stage of the process is logically connected and contributes to transforming unprocessed data into meaningful insights.

The process begins with data collection, where the dataset is imported into Power BI. This is followed by data cleaning and preprocessing using Power Query to ensure accuracy and consistency. After cleaning, the data is organized through data modeling, where relationships between different tables are established to create a structured framework.

Once the data model is prepared, DAX (Data Analysis Expressions) is used to create calculated measures and columns. These calculations enable dynamic analysis and support advanced metrics such as total sales, profit margin, and average discount.

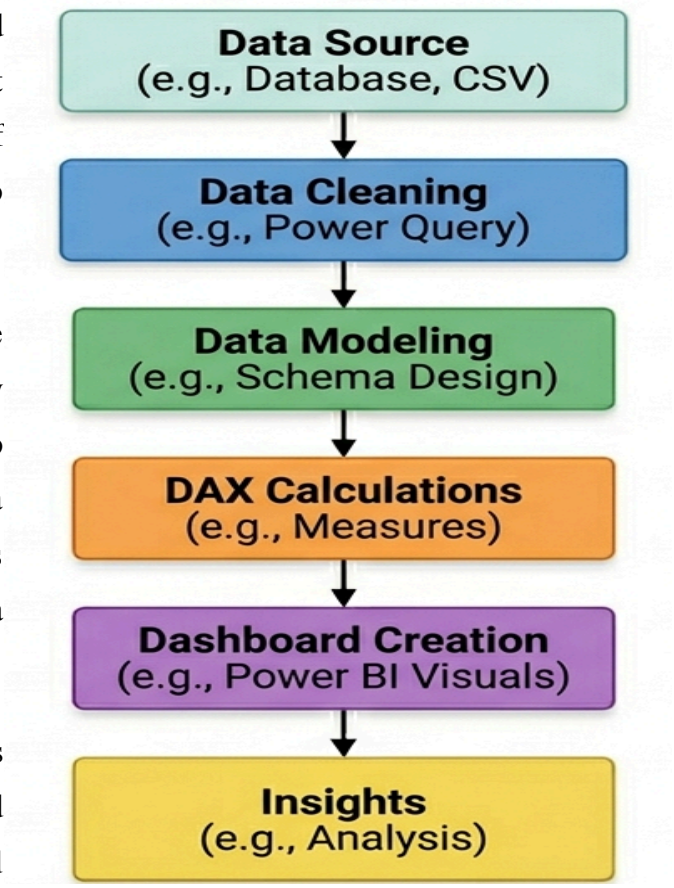

Figure 3.2: Data Analysis & Visualization Process

Finally, the processed data is used to develop interactive dashboards, where various visualizations are created to represent trends, comparisons, and relationships within the data. These dashboards provide a user-friendly interface that allows users to explore information dynamically through filters and slicers. The final stage involves deriving meaningful insights from these visualizations, enabling better understanding of business performance and supporting informed decision-making.

3.6. DAX Measures

DAX (Data Analysis Expressions) is a powerful formula language used in Power BI to create calculated measures and columns that support advanced data analysis. These measures enable dynamic calculations that automatically update based on user interactions such as filters and slicers applied within the dashboard.

In this project, DAX is used to compute key performance indicators that are essential for evaluating business performance. These calculations help in transforming raw data into meaningful metrics that provide deeper insights into sales trends, profitability, and customer behavior.

The use of DAX enhances the analytical capability of the dashboard by enabling real-time computation of values. It allows users to explore data from multiple perspectives and perform comparative analysis across different dimensions such as time, region, and category.

Some key measures created in this project include:
Total Sales, Total Profit, Total Orders, Profit Margin, Average Discount, and Average Order Value (AOV). These measures are fundamental in understanding both overall performance and detailed business insights.

Table 3.2: Key DAX Measures and Their Formulas

Measure Name	Formula	Description
Total Sales	SUM(Sales)	Calculates total revenue
Total Profit	SUM(Profit)	Calculates total earnings
Profit Margin	DIVIDE([Profit],[Sales])	Profit ratio
Avg Discount	AVERAGE(Discount)	Average discount value
Total Orders	COUNT(Order ID)	Number of orders
AOV	DIVIDE([Total Sales],[Total Orders])	Avg order value

3.7. Dashboard Development

The dashboard is developed using Microsoft Power BI to transform processed data into an interactive and visually meaningful representation. The design follows a structured multi-page approach, where each page focuses on a specific analytical objective. This separation ensures clarity, avoids overcrowding of visuals, and allows users to explore different aspects of the data independently.

The dashboard is designed with consistency in layout, color theme, and navigation to enhance user experience. A dark-themed background with contrasting visual elements is used to improve readability and maintain a professional appearance. Interactive slicers are included to enable dynamic filtering of data across all pages, allowing users to customize their analysis based on specific requirements.

Each dashboard page is carefully organized with a clear visual hierarchy, where key performance indicators are placed at the top and detailed charts are positioned below. This structured layout guides users through the information in a logical manner, making it easier to interpret and analyze the data.

Various types of visualizations such as line charts, bar charts, column charts, scatter plots, and donut charts are used to represent different aspects of the data. Each chart is selected based on its suitability for the type of analysis being performed, ensuring effective communication of insights.

Additionally, the dashboard incorporates interactive features such as tooltips, highlighting, and cross-filtering, which allow users to explore relationships within the data more effectively. These features enhance engagement and provide a deeper level of analysis without increasing complexity.

Overall, the dashboard development process focuses on creating a balance between visual appeal and analytical functionality. The final output is an intuitive and user-friendly system that enables users to identify trends, compare performance, and make informed decisions efficiently.

3.7.1 Page 1: Overview Dashboard

The Overview Dashboard serves as the entry point of the system, providing a high-level summary of business performance. It is designed to present key performance indicators (KPIs) and general trends in a concise and easily interpretable format.

Key Components Used

- **KPI Cards**

Figure 3.3: KPI Cards displaying key performance indicators of sales, profit, orders, margin, discount

KPI (Key Performance Indicator) cards are used to present essential business metrics such as total sales, total profit, total orders, profit margin, and average discount in a compact and visually prominent format. These cards are placed at the top of the dashboard to ensure immediate visibility and quick understanding of overall performance.

KPI cards play a crucial role in summarizing large volumes of data into a few key figures that are easy to interpret. They allow users to quickly assess whether the business is performing well or requires attention, without navigating through detailed reports. The use of concise numerical representation reduces cognitive effort and enhances decision-making efficiency.

Reason for selection:

KPI cards are selected because they provide a clear and immediate snapshot of critical business indicators. Their visual prominence ensures that important information is not overlooked. They are particularly useful for managers and stakeholders who require quick insights for strategic decisions.

Alternative Options:

Alternative methods such as tables or text-based summaries could have been used to present these metrics. However, such approaches are less visually engaging and require more effort to interpret. KPI cards, on the other hand, emphasize key values through size, placement, and formatting, making them more effective for dashboard design.

- **Line Chart (Sales Trend Over Time)**

Figure 3.4: Line Chart representing monthly sales trend over time

A line chart is used to visualize the trend of sales over a period of time, typically across months. This chart connects data points with a continuous line, making it easier to observe fluctuations, growth patterns, and seasonal variations in sales performance.

The use of a time-based axis allows users to track how sales evolve over different periods, helping in identifying peak seasons, declining trends, or sudden changes in business activity. This information is valuable for forecasting and planning future strategies.

Reason for Selection:

Line charts are specifically designed for time-series analysis and are highly effective in showing continuity and progression. They clearly represent upward and downward trends, making them ideal for analyzing performance over time.

Alternative Options:

A column or bar chart could also be used to display sales over time. However, these charts focus on discrete comparisons rather than continuous trends. As a result, they may not effectively convey the smooth progression or fluctuation of data across time intervals.

- **Bar Chart (Sales by Category)**

Figure 3.5: Bar Chart showing sales distribution across product categories

A horizontal bar chart is used to compare sales across different product categories. Each bar represents a category, and its length corresponds to the total sales value, allowing easy comparison between categories.

This visualization helps in identifying which categories contribute the most to overall revenue and which ones underperform. It also enables quick recognition of differences in sales distribution across categories.

The horizontal orientation is particularly useful when category names are long, as it prevents overlapping of labels and improves readability.

Reason for Selection:

Bar charts are highly effective for comparing values across distinct categories. They provide a clear and direct visual comparison, making it easy to identify the highest and lowest performers. The horizontal layout further enhances readability and ensures a clean presentation.

Alternative Options:

A pie or donut chart could be used to show the proportion of sales by category. However, these charts focus on percentage distribution rather than precise comparison, making it difficult to analyze small differences between categories. Therefore, a bar chart is preferred for its clarity and accuracy.

- **Slicers (Year and Category)**

Figure 3.6: Interactive slicers for filtering data

Slicers are added to the dashboard to enable users to filter data dynamically based on specific criteria such as year and product category. These interactive elements allow users to customize their view of the data and focus on particular segments without modifying the underlying dataset. By selecting different values in the slicers, users can instantly update all visuals on the dashboard, making the analysis more flexible and user-driven.

Slicers play a significant role in enhancing the interactivity of the dashboard. They provide a clear and accessible way to apply filters, allowing users to explore trends, compare performance across different time periods, and analyze category-wise variations. This feature supports deeper analysis by enabling users to view data from multiple perspectives.

Reason for Selection:

Slicers are chosen because they offer an intuitive and visually accessible method for filtering data. They are easy to use, clearly visible on the dashboard, and provide immediate feedback by dynamically updating all connected visuals. This makes them highly effective for interactive data exploration.

Alternative Options:

The filters pane could also be used to apply filters; however, it is less visible and requires additional navigation. Slicers, on the other hand, are directly placed on the dashboard, making them more user-friendly and convenient for real-time interaction.

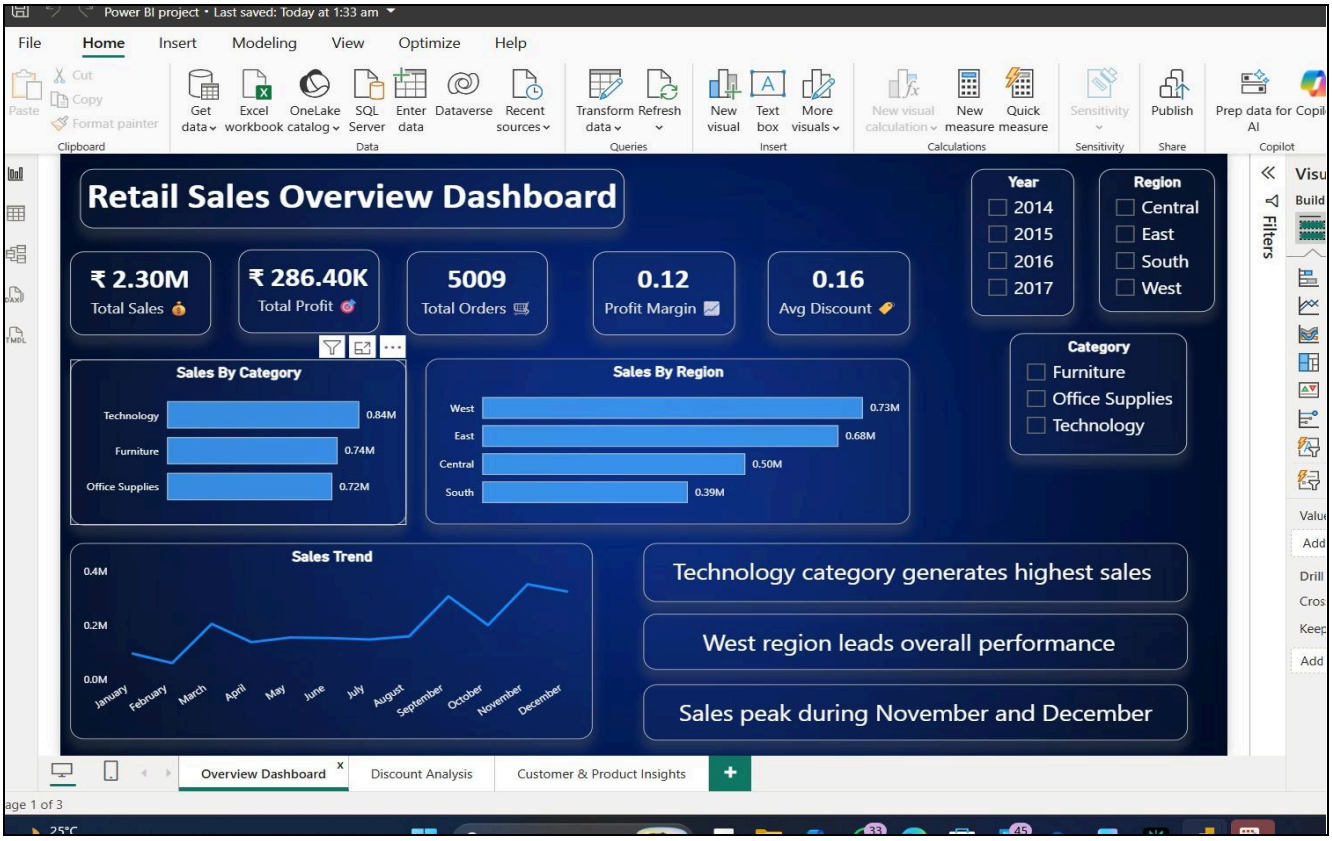

Figure 3.7: Overview Dashboard Layout

3.7.2 Page 2: Discount Analysis

This page is specifically designed to analyze the impact of discount strategies on overall business performance. The primary objective is to examine how varying levels of discounts influence both sales volume and profitability. Understanding this relationship is critical for businesses, as discounts can attract customers but may also reduce profit margins if not managed effectively.

The page combines multiple visualization techniques to provide a comprehensive view of discount behavior, enabling users to identify patterns, correlations, and critical thresholds where profitability is affected. By integrating different charts, the dashboard allows users to observe both detailed and aggregated perspectives of discount impact. This dual-level analysis helps in understanding not only overall trends but also individual variations within the dataset.

In addition, this page supports comparative analysis by allowing users to evaluate how different discount ranges perform across categories and regions. This helps in identifying which segments respond positively to discounts and which may lead to losses. The visualizations also assist in detecting patterns such as diminishing returns, where increasing discounts no longer result in proportional growth in sales.

The analysis further enables businesses to assess the effectiveness of their pricing strategies by highlighting the trade-off between revenue growth and profit sustainability. It helps in identifying optimal discount ranges that maximize sales without significantly affecting profitability. Such insights are essential for designing balanced promotional strategies that align with long-term business goals.

Moreover, the interactive nature of the dashboard allows users to experiment with different filters and scenarios, making it possible to explore the impact of discounts under various conditions. This enhances decision-making by providing a flexible and user-driven analytical environment.

Overall, this section plays a crucial role in understanding pricing strategies and optimizing discount policies. It provides valuable insights that help businesses strike a balance between increasing customer demand and maintaining sustainable profitability, ultimately supporting data-driven and strategic decision-making.

Key Components Used

- **Scatter Plot (Discount vs Profit)**

Figure 3.8: Scatter plot showing relationship between discount and profit

A scatter plot is used to represent the relationship between discount values and profit at the transaction level. Each point in the plot corresponds to an individual data entry, allowing a detailed and granular view of how profit varies with different discount values. This visualization captures the variability in the data and helps in understanding the distribution of profit across different discount levels.

The scatter plot helps in identifying important patterns such as whether higher discounts consistently lead to lower profits or if there are certain cases where discounts still result in positive profitability. It also reveals clusters of data points, indicating common trends, and highlights outliers that may represent unusual or exceptional business scenarios. These insights are valuable for detecting anomalies and understanding real-world variations in sales performance.

In addition, the scatter plot allows users to visually assess the strength and direction of the relationship between discount and profit. It helps in determining whether the relationship is linear, non-linear, or inconsistent across different ranges of discount values. This level of detail is not easily achievable through aggregated visualizations.

Reason for Selection:

Scatter plots are highly effective for analyzing relationships between two continuous variables. They provide a detailed view of data distribution and enable users to identify correlations, clusters, and outliers. This makes them particularly useful for exploring complex relationships that are not immediately visible in summarized data.

Alternative Options:

A line chart or bar chart could be used to display aggregated trends; however, they do not represent individual data points and may hide important variations within the dataset. These charts focus on overall trends rather than detailed relationships. Therefore, a scatter plot is more suitable for in-depth analysis and accurate interpretation of the relationship between discount and profit.

- **Column Chart (Sales by Discount Category)**

Figure 3.9: Column chart showing sales across discount categories

A column chart is used to compare total sales across different discount categories such as low, medium, and high. The categorization of discount values into defined groups simplifies complex numerical data and makes it easier to interpret patterns in sales behavior. By grouping discounts into meaningful ranges, the chart transforms raw data into an easily understandable format, enabling users to quickly analyze how different discount levels influence overall sales performance.

This visualization helps in identifying whether higher discounts lead to increased sales volume or if moderate discounts are more effective in driving revenue. It allows users to observe how customer purchasing behavior changes with varying discount levels, providing insights into price sensitivity and demand patterns. Such analysis is valuable for evaluating promotional strategies and determining the most effective discount range for maximizing revenue.

The column chart also enables clear comparison between categories, allowing users to determine which discount range contributes the most to total sales. Differences in column heights make it easy to identify dominant categories and recognize trends at a glance. This visual clarity supports faster decision-making and reduces the need for detailed numerical analysis.

In addition, the vertical structure of the column chart allows users to easily observe differences in magnitude, making it simple to compare values across categories. The inclusion of data labels further enhances readability by displaying exact numerical values alongside the visual representation. This combination of visual and numerical information improves both accuracy and interpretability.

Furthermore, the chart supports business analysis by highlighting whether increasing discounts consistently results in higher sales or if there is a point of diminishing returns. This insight is essential for developing effective pricing and discount strategies that balance sales growth with profitability.

Reason for Selection:

Column charts are highly effective for comparing aggregated values across distinct categories. They provide a clear visual distinction between groups and allow users to quickly identify trends, differences, and patterns. Their vertical structure makes it easy to observe variations in magnitude, helping users interpret data at a glance. Additionally, their straightforward and familiar design ensures that even non-technical users can easily understand the information presented without requiring detailed analysis.

Alternative Options:

A table could present the same information in a numerical format; however, it lacks visual impact and makes comparison less intuitive. A bar chart could also be used as an alternative, but a column chart is preferred due to its suitability for vertical comparison and its alignment with common visualization practices in dashboards. This makes the analysis more structured and visually appealing.

- **Slicers (Category and Region)**

Figure 3.10: Slicers for filtering data by year, region, and category

Interactive slicers are incorporated to allow users to filter data based on product category and geographical region. These slicers enable focused and customized analysis by allowing users to select specific categories or regions and instantly view the corresponding changes across all visuals on the dashboard. This functionality helps in understanding how discount impact varies across different segments and supports more targeted insights.

Slicers improve the analytical capability of the dashboard by allowing users to explore multiple scenarios without altering the original dataset. For example, users can compare how discounts affect profit in one region versus another or analyze category-specific performance under different discount levels. This flexibility enhances decision-making by providing a more detailed and segmented view of the data.

Reason for Selection:

Slicers enhance user interactivity and provide flexibility in data exploration. They allow users to customize their analysis in real-time and observe immediate changes in the visuals. Their visual presence on the dashboard makes them easy to use and improves overall user experience.

Alternative Options:

Filter panels could perform a similar function; however, they are less visible and require additional steps to access. Slicers, on the other hand, are directly available on the dashboard, making them more intuitive, accessible, and user-friendly.

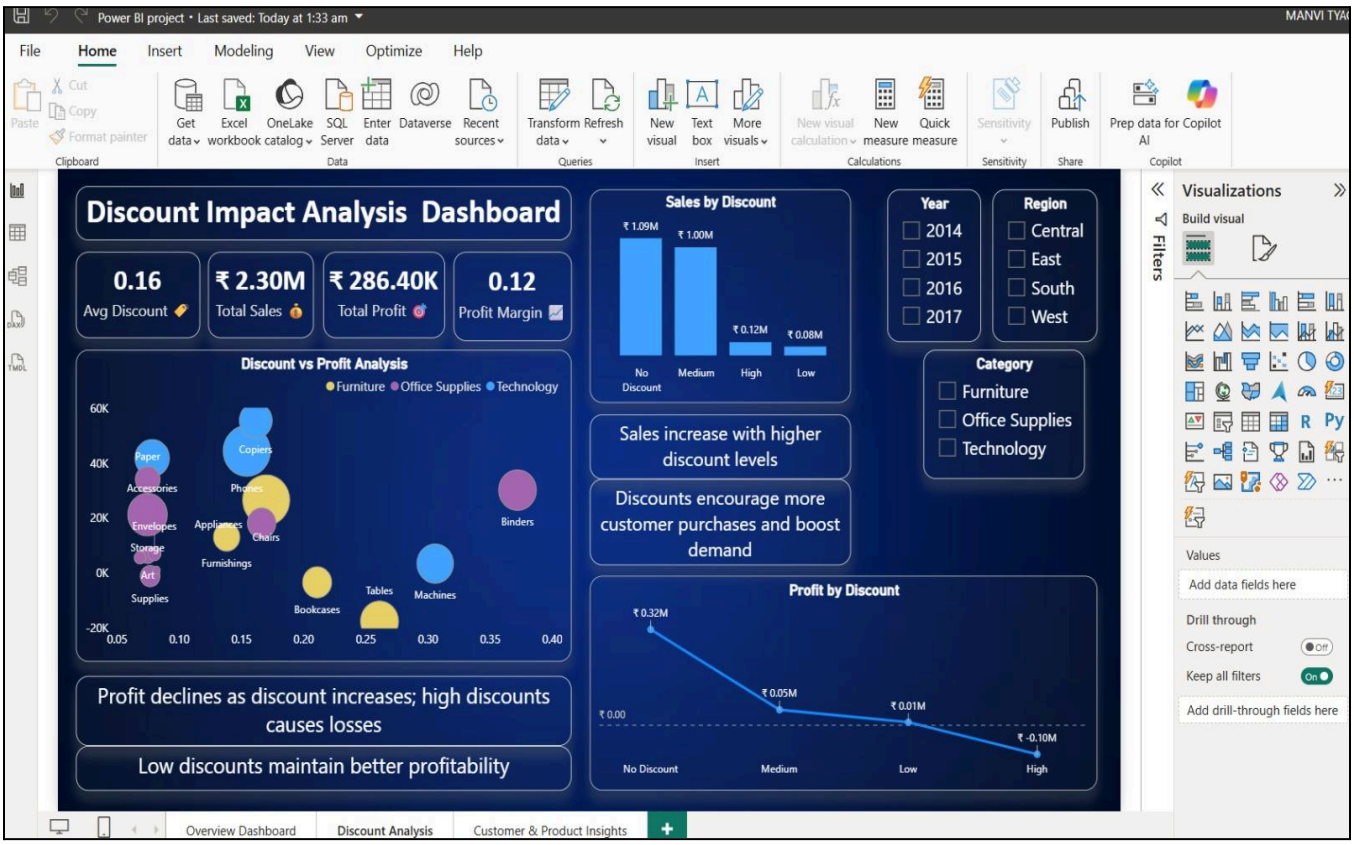

Figure 3.11: Discount Analysis Dashboard Layout

3.7.3 Page 3: Customer & Product Insights

This page is specifically designed to analyze the key contributors to overall business performance by focusing on customers, products, and market segments. Unlike the overview dashboard, which presents a general summary, this page provides a deeper analytical perspective by identifying the core drivers of revenue and profitability.

The primary objective of this page is to support strategic decision-making by highlighting high-value customers, top-performing products, and dominant customer segments. By isolating these critical components, the dashboard enables businesses to focus on areas that generate the highest returns and optimize their strategies accordingly.

This page also helps in understanding the distribution of revenue across different entities, revealing whether business performance is evenly distributed or concentrated among a few key contributors. Such insights are essential for improving customer retention, product positioning, and market targeting.

Key Components Used

- **KPI Cards (Customer & Product Metrics)**

Figure 3.12: KPI cards showing customer and product performance metrics

KPI cards are used to display essential metrics such as total number of customers, average order value, top customer sales, and top product sales. These metrics are carefully selected to shift the focus from overall performance to performance drivers.

These indicators provide insights into customer purchasing behavior, spending patterns, and product contribution to revenue. For instance, average order value reflects customer spending efficiency, while top customer and product metrics highlight key revenue generators.

The placement of KPI cards at the top ensures immediate visibility and allows users to quickly grasp the most important aspects of customer and product performance.

Reason for Selection:

KPI cards are chosen for their ability to present critical information in a concise and visually prominent format. They enable quick interpretation and are particularly useful for decision-makers who require immediate insights without navigating through detailed data.

- **Table (Top Customers)**

Top 10 Customers by Sales

Customer Name	Total Sales
Sean Miller	₹ 25.04K
Tamara Chand	₹ 19.05K
Raymond Buch	₹ 15.12K
Tom Ashbrook	₹ 14.60K
Adrian Barton	₹ 14.47K
Ken Lonsdale	₹ 14.18K
Sanjit Chand	₹ 14.14K
Hunter Lopez	₹ 12.87K
Sanjit Engle	₹ 12.21K
Christopher Conant	₹ 12.13K

Figure 3.13: Table showing top 10 customers by sales

A table is used to display the top customers based on total sales, along with their ranking. This visualization provides a detailed and precise representation of customer contributions to overall revenue.

The inclusion of ranking allows users to easily identify the most valuable customers and understand the concentration of sales among them. It also helps in recognizing patterns such as whether a small number of customers contribute a significant portion of total revenue.

Additionally, tables allow for exact value comparison, which is essential when analyzing financial data. This level of detail supports informed decision-making, particularly in areas such as customer relationship management and targeted marketing strategies.

Reason for Selection:

Tables are ideal for presenting detailed numerical information and exact values in a structured format. They provide clarity, precision, and easy readability, making them suitable for scenarios where accurate comparison is required. Tables also allow users to view multiple attributes simultaneously, such as customer names, sales values, and rankings, which enhances analytical understanding.

Alternative Options:

A bar chart could visually represent customer rankings and provide a quick comparison. However, it may not display precise numerical values as clearly as a table. Additionally, charts focus more on visual trends rather than exact figures. Therefore, a table is preferred for its ability to present detailed and accurate information effectively.

- **Bar Chart (Top Products)**

Figure 3.14: Bar chart showing top 5 products by sales

A horizontal bar chart is used to display the top-performing products based on sales. This visualization highlights the products that contribute the most to overall revenue, enabling users to identify key revenue drivers and understand product-level performance. By focusing on top products, the chart helps in recognizing which items are most popular and generate the highest demand in the market.

The horizontal orientation is particularly beneficial when dealing with long product names, as it improves readability and prevents label overlap. This ensures that the chart remains clear, well-structured, and easy to interpret without compromising the visibility of textual information.

In addition, the chart supports quick comparison between products by clearly showing differences in sales values through bar lengths. This allows users to easily identify performance gaps, detect underperforming products, and explore opportunities for improvement or strategic focus. It also assists in decision-making related to inventory management, marketing, and product promotion.

The inclusion of data labels further enhances the visualization by providing exact numerical values alongside the bars, combining both visual comparison and precise information in a single view.

Reason for Selection:

Bar charts are highly effective for ranking and comparison, especially when dealing with categorical data. The horizontal layout enhances readability and provides a clean and organized presentation, making it easier for users to analyze product performance at a glance. It is particularly suitable when category labels are long or complex, as it prevents text overlap and ensures clear visibility of labels.

Additionally, bar charts allow users to quickly identify top-performing and low-performing items by comparing the lengths of the bars. This makes it easier to detect performance gaps and trends without requiring detailed numerical analysis. The visual clarity and simplicity of bar charts make them an ideal choice for presenting ranked data in a structured and user-friendly manner.

Alternative Options:

A vertical column chart could also be used to represent product performance. However, when product names are lengthy, it may lead to overlapping labels and reduced clarity. Additionally, vertical charts may require more space for proper display. Therefore, a horizontal bar chart is preferred as it offers better readability, efficient use of space, and improved visual clarity. Moreover, horizontal bar charts allow for easier comparison of values, as the length of each bar can be quickly interpreted without visual strain. This makes the analysis more intuitive and enhances the overall user experience.

- **Donut Chart (Profit by Category)**

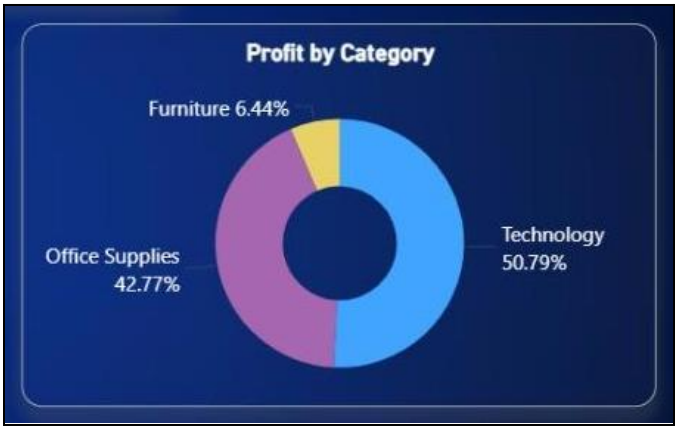

Figure 3.15: Donut chart showing profit distribution by category

A donut chart is used to represent the proportion of profit contributed by each product category. This visualization provides a clear understanding of how profit is distributed across categories such as technology, furniture, and office supplies.

By focusing on proportions rather than absolute values, the chart helps in identifying which categories are most profitable and which require improvement. It also highlights the imbalance, if any, in profit contribution.

The donut structure, with its central space, enhances visual appeal and allows for better presentation of percentage values.

Reason for Selection:

Donut charts are effective for displaying proportional data and are visually engaging. They simplify complex data into easily understandable segments, making them suitable for summarizing distribution patterns.

Alternative Options:

A bar chart could display profit values, but it does not emphasize proportional relationships as effectively as a donut chart. Therefore, the donut chart is preferred for this purpose.

- **Segment Analysis Chart**

Figure 3.16: sales distribution by customer segment

This chart is used to analyze sales distribution across different customer segments such as consumer, corporate, and home office. It provides insights into which segment contributes the most to overall revenue.

Understanding segment performance is crucial for identifying target markets and designing segment-specific strategies. For example, if one segment dominates sales, businesses can focus on strengthening relationships with that segment while also exploring opportunities in underperforming segments.

Reason for Selection:

Segment analysis helps in understanding customer diversity and market distribution. It supports strategic planning by identifying key customer groups and their contribution to business performance.

- **Insights Section**

Figure 3.17: Key insights derived from customer and product analysis

Text-based insight boxes summarize key findings and highlight important patterns, making the dashboard easier to understand and more effective for decision-making.

Reason for Selection:

Including insights improves usability by presenting conclusions clearly, enabling users to understand results and make informed decisions.

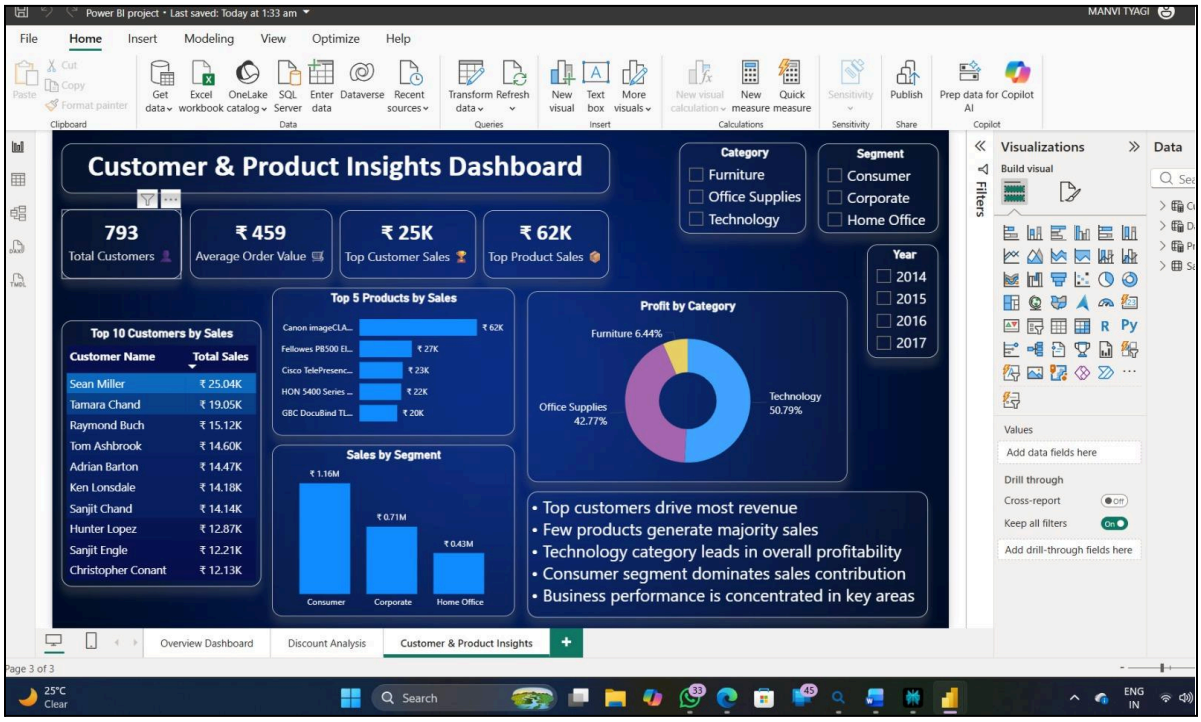

Figure 3.18: Customer & Product Dashboard Layout

3.7.4 Design Considerations

The dashboard design is developed by following several key principles to ensure clarity, usability, consistency, and visual effectiveness. These design considerations play a crucial role in enhancing user experience and ensuring that the information is communicated in an intuitive and meaningful way.

A consistent color scheme is maintained across all pages to create a unified and professional appearance. The use of a dark background combined with contrasting colors for visuals improves readability and helps in highlighting important data points. Consistency in colors also ensures that users can easily associate similar elements across different pages.

Proper alignment and spacing of visuals are carefully maintained to avoid clutter and ensure a balanced layout. Each visual is positioned in a structured manner, allowing users to navigate the dashboard smoothly without confusion. Adequate spacing between elements improves visual clarity and prevents overcrowding of information.

Appropriate chart types are selected based on the nature of the data and the type of analysis required. For example, line charts are used for trends, bar charts for comparisons, and scatter plots for relationship analysis. This ensures that the data is represented in the most effective and interpretable form.

The number of visuals on each page is intentionally limited to maintain simplicity and avoid information overload. Instead of adding excessive charts, only the most relevant visuals are included to focus on key insights. This approach improves user understanding and reduces cognitive load.

Interactive elements such as slicers are incorporated to allow users to filter and explore data dynamically. These features enhance user engagement and provide flexibility in analysis, enabling users to view data from different perspectives without modifying the dashboard structure.

Special attention is given to readability by using appropriate font sizes, clear labels, and proper contrast between text and background. This ensures that all information is easily visible and accessible, even for non-technical users.

Additionally, design consistency is maintained in terms of layout, font style, and visual hierarchy. Important elements such as KPIs are placed at the top, while detailed charts are positioned below, guiding the user's attention in a logical flow.

Another important consideration in the dashboard design is the use of visual hierarchy to guide user attention effectively. Key elements such as KPI cards are placed prominently at the top, followed by detailed charts and supporting visuals below. This structured arrangement ensures that users first observe the most critical information before exploring detailed analysis. The logical flow of information enhances navigation, reduces confusion, and improves overall user experience by making the dashboard intuitive and easy to interpret.

3.7.5 Summary

The dashboard is designed to provide a comprehensive and user-friendly analytical experience by combining simplicity with functionality. Each page is structured to address a specific aspect of the analysis, ensuring that users can focus on one area at a time without being overwhelmed by information.

The use of appropriate visualization techniques ensures that complex data is presented in an easy-to-understand format. Interactive features such as slicers and filters further enhance the usability of the dashboard by allowing users to customize their analysis based on specific requirements.

The structured layout, consistent design, and careful selection of visuals collectively contribute to an effective communication of insights. Users can quickly identify trends, compare performance, and derive meaningful conclusions from the data.

Overall, the dashboard successfully transforms raw data into actionable insights by providing a balance between visual appeal and analytical depth. It is designed to cater to both technical and non-technical users, making it a practical and efficient tool for data-driven decision-making.

4. RESULTS ANALYSIS AND VALIDATION

4.1. Overview

This chapter presents the results obtained from the developed Power BI dashboard and provides a detailed analysis of retail sales performance. The results are visualized through interactive dashboards, enabling better understanding of trends, patterns, and relationships within the data. The use of visual analytics allows complex datasets to be interpreted in a simplified and structured manner.

The analysis is divided into three sections based on the dashboard pages: Overview, Discount Analysis, and Customer & Product Insights. Each section highlights key findings and their implications for business decision-making. This structured approach ensures that both high-level summaries and detailed insights are effectively communicated.

The chapter focuses not only on presenting the results but also on interpreting them in a meaningful way. By examining patterns across different dimensions such as time, category, region, and customer segments, the analysis provides a comprehensive understanding of business performance.

Furthermore, the insights derived from the dashboards help in identifying strengths, weaknesses, and areas for improvement. These findings can be used by businesses to optimize strategies, improve operational efficiency, and make data-driven decisions.

Overall, this chapter plays a crucial role in validating the effectiveness of the developed dashboard and demonstrating how data visualization can support strategic planning and performance evaluation.

In addition, the interactive nature of the dashboard allows users to explore the data dynamically by applying filters and slicers. This enables a more personalized analysis, where users can focus on specific categories, regions, or time periods to gain deeper insights. Such flexibility enhances the overall effectiveness of the analysis and ensures that the results are relevant to different business scenarios.

4.2. Overview Dashboard Analysis

Figure 4.1: Overview Dashboard showing key performance indicators and sales trends

The Overview Dashboard provides a comprehensive summary of business performance through key performance indicators and visualizations. It serves as the starting point for analysis and gives a quick understanding of the overall state of the business. By combining multiple visual elements in a single view, the dashboard allows users to quickly grasp both high-level performance and underlying trends.

The KPI cards display important metrics such as total sales, total profit, total orders, profit margin, and average discount. These indicators provide an immediate snapshot of business performance and help in evaluating overall efficiency. They also enable quick comparison between different metrics, allowing users to assess whether increased sales are translating into higher profitability or being affected by discount strategies.

The sales trend line chart shows how sales vary over time, enabling the identification of seasonal patterns and growth trends. It highlights periods of high and low performance, which can be useful for planning and forecasting. The continuous nature of the chart helps in understanding the direction of business growth and detecting any sudden fluctuations that may require further investigation.

The category-wise and region-wise charts provide insights into how sales are distributed across different segments. These visuals help in identifying high-performing categories and regions, as well as areas that require improvement. By comparing these segments, businesses can allocate resources more effectively and focus on regions or categories that offer higher returns.

In addition, the overview dashboard enables users to quickly compare multiple performance indicators within a single view, reducing the need to navigate across different reports. The integration of various visuals allows for a holistic understanding of business operations, where trends, distributions, and key metrics can be analyzed simultaneously. This comprehensive perspective supports faster decision-making and helps stakeholders identify opportunities for improvement and areas requiring immediate attention.

Furthermore, the dashboard supports interactive analysis through features such as slicers and cross-filtering, allowing users to explore specific time periods, categories, or regions. This interactivity enhances the analytical depth by enabling users to drill down into the data and uncover detailed insights beyond the surface-level summary.

The structured layout and clear visual hierarchy guide users from key metrics to detailed analysis, ensuring a smooth flow of information. This design approach improves usability and ensures that important insights are not overlooked.

Key Insights

- Sales performance shows variation over time, indicating the presence of seasonal trends.
- Certain product categories contribute more significantly to total revenue.
- Regional analysis reveals differences in performance across locations.

4.3. Discount Analysis

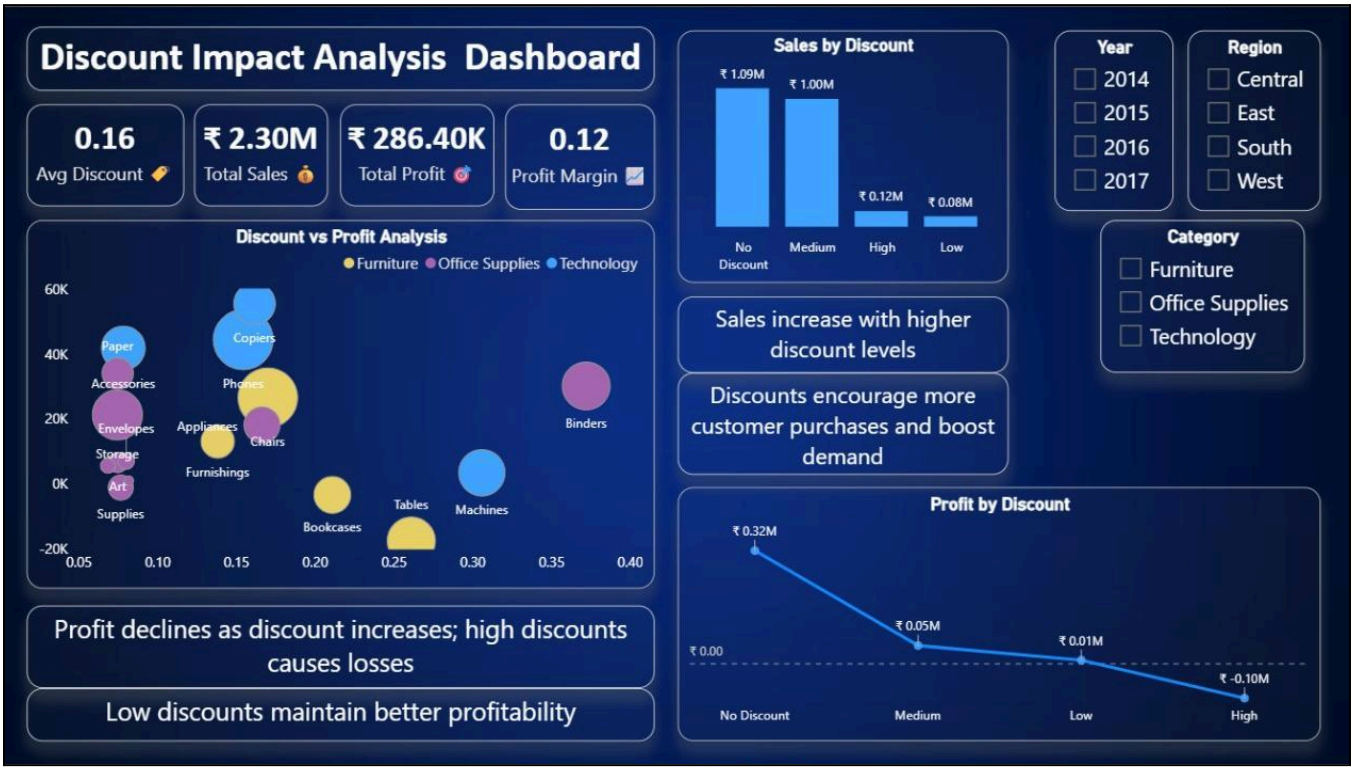

Figure 4.2: Discount Analysis Dashboard illustrating the impact of discounts on sales and profit

The Discount Analysis Dashboard is designed to evaluate the impact of discount strategies on both sales performance and profitability. Discounts play a crucial role in attracting customers and increasing sales volume; however, their effect on profit margins must be carefully analyzed. This section provides a detailed examination of how varying discount levels influence business outcomes, helping in identifying optimal pricing strategies.

The scatter plot illustrates the relationship between discount and profit at a detailed level, where each point represents an individual transaction. This visualization enables the identification of patterns, clusters, and outliers, providing a deeper understanding of how profit behaves across different discount values. It clearly shows that as discount levels increase, profit tends to decline, indicating a negative correlation between the two variables.

The column chart representing sales by discount category groups discount values into categories such as low, medium, and high. This grouping simplifies the analysis and makes it easier to compare how sales volume changes across different discount levels. The chart reveals that higher discounts often lead to increased sales, as customers are more likely to make purchases when offered price reductions.

The line chart for profit by discount highlights the trend of profitability as discount levels increase. The inclusion of a zero reference line provides a clear distinction between profit and loss regions, making it easier to identify the threshold beyond which discounts become unprofitable. This chart effectively demonstrates how excessive discounting can lead to negative financial outcomes.

Additionally, the combination of multiple visualizations on this page allows for cross-analysis, where users can simultaneously observe the impact of discounts on both sales and profit. This integrated approach enhances understanding and supports more informed decision-making.

Furthermore, the analysis highlights the importance of maintaining an optimal discount strategy that balances customer attraction and profitability. While discounts can be an effective tool for boosting sales, excessive reliance on high discount levels may reduce overall business sustainability. By observing the patterns across different visuals, businesses can identify the most effective discount range that maximizes revenue without significantly impacting profit margins. This insight is valuable for developing data-driven pricing strategies and improving overall financial performance.

Key Insights:

- Sales tend to increase with higher discount levels, indicating customer sensitivity to pricing.
- Profit decreases as discount levels rise, reflecting reduced profit margins.
- High discount ranges often result in negative profitability, highlighting financial risk.
- Moderate discount levels provide a balance between increasing sales and maintaining profit.
- Excessive discounting may lead to short-term gains in sales but long-term losses in profitability.

4.4. Customer & Product Insights

Figure 4.3: Customer & Product Insights Dashboard highlighting top contributors & performance drivers

This section focuses on identifying the key contributors to overall business performance by analyzing customers, products, and segments. It provides a deeper understanding of how revenue is distributed and highlights the primary drivers that influence sales and profitability. By examining these elements, businesses can identify where their strengths lie and where improvements are needed. This level of analysis helps in uncovering patterns that are not visible in high-level summaries.

The KPI cards highlight important metrics such as total customers, average order value, and top-performing customers and products. These metrics provide insights into customer purchasing behavior, spending patterns, and product performance. For instance, average order value reflects how much customers spend per transaction, while top customer and product metrics help in identifying major revenue contributors. These indicators also assist in evaluating customer value and understanding purchasing efficiency.

The table of top customers shows that a small group of customers contributes a significant portion of total sales. This indicates a high concentration of revenue among key customers, emphasizing the importance of maintaining strong relationships with these high-value clients. It also suggests that customer retention strategies, loyalty programs, and personalized marketing efforts can significantly impact overall business performance.

The bar chart of top products highlights the products that generate the highest sales, enabling businesses to identify best-selling items. This information is useful for inventory management, promotional planning, and product positioning. It also helps in recognizing which products require additional focus to improve performance and which ones can be leveraged for maximizing revenue through targeted promotions.

The donut chart illustrates the distribution of profit across different product categories, providing a clear view of which categories contribute the most to profitability. This helps businesses understand which areas are most profitable and where resources should be allocated for maximum returns. It also highlights any imbalance in profit contribution, which may indicate the need for strategic adjustments in pricing or cost management.

The segment analysis chart provides insights into how different customer segments contribute to overall sales. It highlights the dominance of certain segments and helps in identifying target markets for future growth. This analysis supports strategic decisions related to marketing, customer engagement, and business expansion by identifying which segments offer the highest potential.

Furthermore, the analysis reveals the interdependence between customers and products, showing how specific products are favored by certain customer groups. This insight can be used to design targeted campaigns and improve customer satisfaction by aligning product offerings with customer preferences.

In addition, the combined use of these visualizations allows for a comprehensive analysis of business performance from multiple perspectives. By integrating customer, product, and segment-level insights, the dashboard provides a holistic view that supports informed decision-making and strategic planning. It enables businesses to move beyond descriptive analysis and adopt a more strategic and focused approach to performance improvement.

Key Insights

A small number of customers contribute a large share of total revenue, indicating revenue concentration.

- Top-performing products act as major drivers of overall sales performance.
- Profit contribution varies significantly across categories, highlighting differences in profitability.
- Consumer segment contributes the highest share of sales, making it the dominant customer group.
- Business performance is influenced by a limited set of key customers and products.
- Customer behavior and product performance are closely linked, impacting overall business outcomes.

4.5. Validation of Results

The results obtained from the dashboard are validated through consistency checks and logical analysis. The relationships observed between variables, such as the inverse relationship between discount and profit, align with expected business behavior. This alignment confirms that the analytical outputs are meaningful and reflect real-world business dynamics.

The use of multiple visualizations ensures that findings are supported by different perspectives, increasing the reliability of the analysis. For example, both scatter and line charts confirm the negative impact of high discounts on profitability, reinforcing the consistency of the observed pattern. This cross-validation across visuals strengthens confidence in the results.

Additionally, the consistency of insights across different filters and slicers further validates the accuracy of the results. The dashboard allows dynamic interaction, enabling users to verify patterns across various categories and regions. This ensures that the conclusions remain stable even when the data is viewed from different perspectives.

Furthermore, the validation process is strengthened by cross-verifying key metrics such as total sales, profit, and order counts across different visuals and calculations. The consistency of these values ensures that the data transformation and DAX measures are correctly implemented. This multi-level validation approach enhances the credibility of the dashboard and confirms that the insights generated are accurate, reliable, and suitable for supporting data-driven business decisions.

In addition, the validation process includes checking for data completeness and ensuring that no significant anomalies or inconsistencies exist within the dataset. Any unusual patterns observed in the visuals are carefully analyzed to determine whether they represent genuine business scenarios or potential data issues.

The use of a structured data model and well-defined relationships also contributes to the accuracy of results, as it ensures that calculations are performed correctly across different tables. This minimizes errors and improves the overall reliability of the analysis. Overall, the validation process confirms that the dashboard provides trustworthy insights and serves as a dependable tool for analyzing retail sales performance and supporting strategic decision-making.

4.6. Summary

This chapter presented the results and analysis derived from the Power BI dashboard. The findings highlight important aspects of retail business performance, including sales trends, discount impact, and key contributors to revenue.

The analysis demonstrates that while discounts can increase sales, they may negatively affect profitability if not managed properly. It also reveals that business performance is often concentrated among specific customers and products.

These insights provide valuable guidance for decision-making and strategic planning, enabling businesses to optimize their operations and improve overall performance.

Furthermore, the results emphasize the importance of using data visualization tools to simplify complex datasets and uncover meaningful patterns. The dashboard not only enhances analytical efficiency but also enables users to identify trends and relationships that support proactive decision-making. By leveraging these insights, businesses can develop more effective strategies, improve resource allocation, and achieve sustainable growth.

5. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This project focused on developing an interactive Power BI dashboard to analyze retail sales performance and understand the impact of discount strategies on business outcomes. The primary objective was to transform raw transactional data into meaningful insights that support data-driven decision-making.

The dashboard successfully integrates multiple analytical perspectives, including overall performance, discount impact, and customer-product insights. Through the use of visualization techniques and calculated measures, the project provides a comprehensive view of business operations.

The analysis reveals that while discounts can significantly increase sales volume, they often lead to a reduction in profit margins when applied excessively. This highlights the importance of maintaining an optimal balance between attracting customers and preserving profitability.

Additionally, the results indicate that business performance is not evenly distributed. A small number of customers and products contribute a large share of total revenue, emphasizing the need for targeted strategies focused on high-value entities.

The use of Power BI proved effective in presenting complex data in a simplified and interactive format. The dashboard enables users to explore data dynamically, identify trends, and derive actionable insights. Overall, the project demonstrates the practical application of business intelligence tools in solving real-world problems.

Furthermore, the project highlights the importance of data-driven decision-making in modern business environments. By leveraging interactive visualizations and analytical tools, organizations can gain deeper insights into their operations and respond more effectively to changing market conditions. The developed dashboard not only improves analytical efficiency but also provides a scalable foundation for future enhancements and advanced analytics.

5.2. Key Findings

The following key findings were derived from the analysis conducted using the Power BI dashboard. These findings provide a clear understanding of retail business performance and highlight important patterns observed across different dimensions of the dataset.

- Sales performance varies over time, indicating the presence of seasonal trends and fluctuations in demand.
- Higher discount levels tend to increase sales volume, as customers are more attracted to lower prices.
- However, increased discounts negatively impact profitability, leading to reduced profit margins.
- Moderate discount strategies provide an optimal balance between generating revenue and maintaining profitability.
- Revenue is concentrated among a small group of customers and products, indicating dependency on key contributors.
- The technology category contributes significantly to overall profit, making it a major driver of business success.

These findings highlight the need for a balanced approach in pricing and marketing strategies. Businesses should focus on optimizing discount levels, strengthening relationships with high-value customers, and promoting top-performing products to maximize overall performance.

Table 5.1: Summary of Key Findings

Aspect	**Observation**
Sales Trend	Seasonal variation observed
Discount Impact	High discount reduces profit
Customer Contribution	Few customers generate most revenues
Product Performance	Top products dominate sales
Segment Contribution	Consumer segment leads

5.3. Advantages of the System

The developed dashboard provides several advantages that enhance business analysis and decision-making capabilities. By integrating data processing and visualization into a single platform, the system offers a streamlined and efficient analytical solution.

- Provides real-time insights, allowing users to monitor trends and make timely decisions.
- Simplifies complex data into easy-to-understand visuals, improving clarity and interpretation.
- Enhances decision-making through interactive analysis, enabling users to explore data dynamically.
- Identifies key performance drivers such as top customers, products, and regions.
- Improves efficiency by reducing manual analysis and saving time.
- Enables quick comparison across different dimensions such as time, category, and region.
- Supports data-driven strategies by providing accurate and reliable insights.

Overall, the system improves analytical efficiency, enhances user experience, and provides a powerful tool for understanding and optimizing business performance.

5.4. Limitations of the System

Despite its advantages, the developed system has certain limitations that may affect the scope and depth of analysis. These limitations highlight areas where further improvements can be made.

- Analysis is limited to historical data only.
- No predictive modeling or forecasting included.
- Results depend on data quality and accuracy.
- Limited to available dataset features.
- External factors are not considered.
- No automated alerts or recommendations.

Overall, while the system provides valuable insights, these limitations indicate opportunities for future enhancement and expansion.

5.5. Future Scope

he project can be further enhanced in several ways to improve its capabilities and applicability. By incorporating additional features and advanced techniques, the system can evolve into a more powerful and comprehensive analytical solution.

Future improvements may include the integration of real-time data sources to enable live monitoring of business performance. This would allow organizations to make faster and more responsive decisions, especially in dynamic market environments where timely insights are critical.

Advanced analytical techniques such as machine learning and predictive modeling can be incorporated to forecast future sales trends and customer behavior. This would provide a proactive approach to decision-making rather than relying solely on historical analysis. Predictive insights can help businesses anticipate demand, optimize pricing strategies, and improve overall planning.

The dashboard can also be expanded to include additional data sources such as inventory management, supply chain data, and customer feedback. This integration would provide a more comprehensive view of business operations and enable better coordination between different functional areas.

Another potential enhancement is the development of a mobile-friendly version of the dashboard, enabling users to access insights on various devices. This would improve accessibility and allow decision-makers to monitor performance anytime and anywhere.

Furthermore, the system can be enhanced by incorporating advanced visualization techniques and customizable dashboards, allowing users to personalize their analysis based on specific business needs. Features such as automated alerts, recommendations, and notifications can also be added to improve responsiveness and usability.

Overall, these enhancements would significantly improve the functionality, scalability, and practical applicability of the system, making it more suitable for real-world business environments.

5.6. Practical Applications

The developed system has several real-world applications across different business domains. It provides valuable insights that can be utilized by organizations to improve performance, optimize strategies, and enhance decision-making processes.

- Optimize pricing and discount strategies.
- Identify high-value customers for retention.
- Monitor performance trends and take actions.
- Improve decision-making using data insights.
- Analyze product performance and manage inventory.
- Design targeted marketing campaigns.
- Allocate resources based on regional performance.

Overall, the system serves as a practical tool for enhancing business intelligence, enabling organizations to make informed decisions, improve efficiency, and achieve better outcomes.

5.7. Summary

This chapter summarized the overall outcomes of the project and highlighted key findings, advantages, and future possibilities. It provided a comprehensive overview of how the developed system contributes to understanding retail business performance.

The developed dashboard demonstrates the effectiveness of data visualization in transforming raw data into actionable insights. The project also provides a strong foundation for further enhancements and real-world applications.

APPENDIX

This section provides supporting information, additional details, and references used in the development of the project. It includes dataset details, tools used, key formulas, and screenshots that support the implementation of the system.

Appendix A: Dataset Information

The dataset used in this project is the Global Superstore dataset obtained from Kaggle. It contains transactional data related to sales, profit, discount, customer details, product categories, and geographical regions. The dataset is structured and suitable for business analysis and visualization.

Source: Kaggle (Global Superstore Dataset)

Appendix B: Tools and Technologies Used

- Microsoft Power BI – Data visualization and dashboard development
- Power Query – Data cleaning and preprocessing
- DAX (Data Analysis Expressions) – Creating calculated measures
- Microsoft Excel/CSV – Data storage and import

Appendix C: Key DAX Measures

- Total Sales = SUM(Sales)
- Total Profit = SUM(Profit)
- Total Orders = COUNT(Order ID)
- Profit Margin = DIVIDE([Total Profit], [Total Sales])
- Avg Discount = AVERAGE(Discount)
- AOV = DIVIDE([Total Sales], [Total Orders])

These measures are used to calculate key performance indicators and support dynamic analysis in the dashboard.

Appendix D: Data Preprocessing Steps

- Removal of duplicate records
- Handling of missing and null values
- Conversion of data types (date, numeric, text)
- Creation of calculated columns (Month, Year, Discount Category)
- Renaming columns for clarity

Appendix E: Dashboard Pages

- Page 1: Overview Dashboard
- Page 2: Discount Analysis
- Page 3: Customer & Product Insights

Each page is designed to focus on a specific aspect of business analysis and provides interactive visualisations.

Appendix F: Assumptions

- Data provided in the dataset is accurate and reliable
- No external factors (market trends, competition) are included
- Analysis is based on historical data only

Appendix G: Abbreviations

- KPI – Key Performance Indicator
- DAX – Data Analysis Expressions
- AOV – Average Order Value
- BI – Business Intelligence

Appendix H: Screenshots

Include the following screenshots:

- **Data Model Diagram**

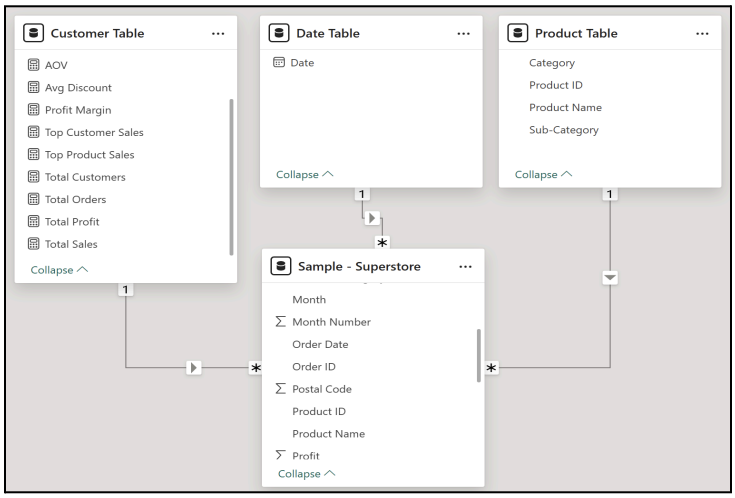

Figure 3.1: Data Model showing relationships between fact and dimension tables

- **Dashboard Page 1**

Figure 4.1: Overview Dashboard showing key performance indicators and sales trends

- **Dashboard Page 2**

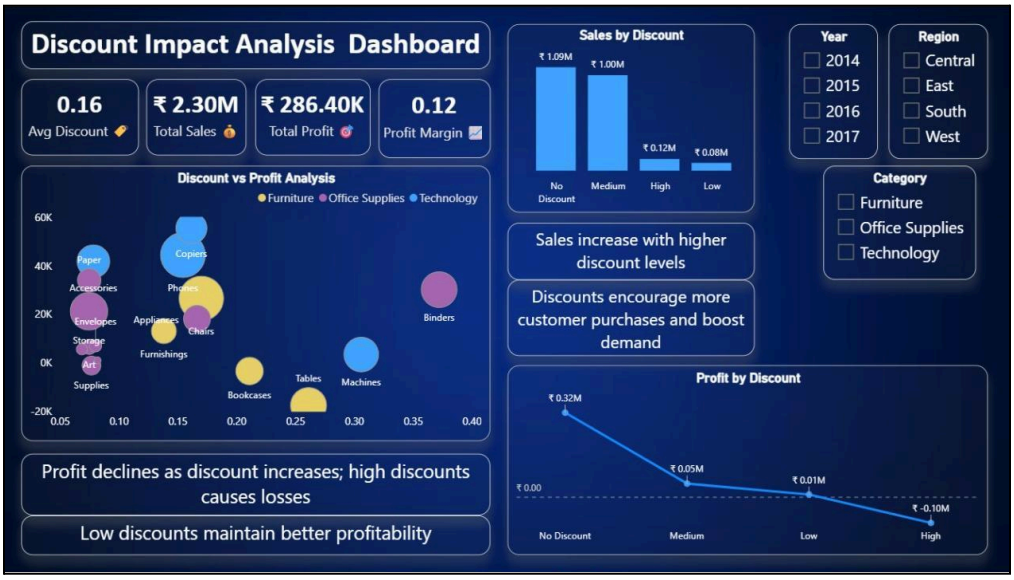

Figure 4.2: Discount Analysis Dashboard illustrating the impact of discounts on sales and profit

- **Dashboard Page 3**

Figure 4.3: Customer & Product Insights Dashboard highlighting top contributors & performance drivers

REFERENCES

The following sources were referred to during the development of this project:

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